Calendar-Arbitrage Prevention

Convex order, integrated quantiles, and a model-agnostic soft hinge

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Abstract

Butterfly-freedom keeps a single smile a valid density; calendar-freedom keeps a *surface* consistent across maturities. This note states precisely what that means and how the Vol-Fitter enforces it. Under fixed-forward normalization with deterministic carry, calendar-freedom is four equivalent statements — normalized calls non-decreasing in maturity, the terminal laws increasing in convex order, the integrated upper-quantile curves ordered, and total implied variance non-decreasing at every fixed log-moneyness — and, by Kellerer's theorem, exactly the condition for the fitted surface to be consistent with *some* martingale. Two enforcements follow. For LQD the integrated upper-quantile curve *is* the asset-share integral every built slice already carries, so the constraint is an elementwise array comparison; expiries are fitted nearest-to-farthest under a soft slack. For the SVI and Multi-Core SIV overlays a *model-agnostic* hinge enforces $w_{\mathrm{far}}(k) \geq w_{\mathrm{near}}(k)$ directly (the local-vol surface needs neither: positivity of its local variance is calendar-cleanliness, Note 04). The note's case file is the *phantom calendar*: a fixed wide floor grid let a steep short expiry's linear-wing extrapolation manufacture violations in a no-data region, flattening far fits live on NVDA and SPY; the confinement fix — sample the floor only on the traded range — is reproduced here with the production calibrator (far-fit error 151 vol bp under the wide grid vs 0.0 under the confined one). A worked example takes an inverted-wing pair from a violation of 0.0059 to 4.6×10^{-7}.

Contents

1 What calendar-freedom means

Assumption 1 (Units and carry)**.** Each expiry is normalized by its own forward: $Y_T = S_T/F_T$ (so $\mathbb{E}[Y_T] = 1$ for every T), quotes live at fixed log-moneyness $k = \log(K/F_T)$, and rates, borrow and dividends are deterministic and absorbed into F_T (Note 06). Comparisons across maturity are made at fixed k — fixed *forward* moneyness, not fixed cash strike. Discrete cash dividends make the cash-strike statement fail while the normalized one survives; stochastic rates would break the equivalence between calendar spreads and the normalized comparison altogether. The event clock of Note 11 reparametrizes maturity within each slice but preserves the *ordering* of expiries, so the statements below are unaffected.

Invariant

1. Where data lives, the fitted surface is increasing in convex order: no calendar spread against it.
2. Enforcement is a *soft* hinge — quotes still win a genuine conflict, and the diagnostic reports what slack was used.
3. `enforceCalendar` off, or no previous slice, is byte-identical to independent fits.
4. The constraint is sampled only where *both* slices are pinned by data — the confinement principle (Note 09).

Theorem 1 (Four faces of calendar-freedom)**.** *Let $\{Y_T\}$ have unit mean, $C_T(k) = \mathbb{E}[(Y_T - e^k)^+]$ the normalized calls, $G_T(\alpha) = \int_\alpha^1 Q_{Y,T}(u)\,\mathrm{d}u$ the integrated upper-quantile curves, and $w_T(k)$ the total implied variances. For $T_i < T_j$ the following are equivalent:*

(i) $C_{T_i}(k) \le C_{T_j}(k)$ *for all k (no calendar-spread arbitrage);*

(ii) $Y_{T_i} \le_{cx} Y_{T_j}$ *(increasing convex order);*

(iii) $G_{T_i}(\alpha) \le G_{T_j}(\alpha)$ *for all $\alpha \in [0,1]$, with equality at $\alpha = 0$;*

(iv) $w_{T_i}(k) \le w_{T_j}(k)$ *for all k.*

Proof sketch. (i)⇔(ii): call payoffs $(y - e^k)^+$ span the convex functions of y (every convex function with equal means integrates against the second derivative of the call in strike — the Breeden–Litzenberger pairing), so ordering all calls with equal means is exactly convex order [1, 3]. (ii)⇔(iii): the Hardy–Littlewood characterization — $\mathbb{E}\phi(Y_i) \le \mathbb{E}\phi(Y_j)$ for all convex ϕ iff the integrated quantile functions are ordered; G is the upper-tail version, and $G(0) = \mathbb{E}Y = 1$ pins the means. (i)⇔(iv): the normalized Black price is strictly increasing in total variance at fixed k, so ordering prices at every k is ordering the implied w at every k [4]. □

Remark 1 (Kellerer: why this is the whole story)**.** If in addition each slice is butterfly-free (a valid density — Notes 01–04), then (ii) for all pairs makes the family $\{\mathrm{law}(Y_T)\}$ a *peacock*, and Kellerer's theorem supplies a martingale with exactly these marginals [2]. So butterfly-freedom per slice plus calendar-freedom across slices is not merely "no static arbitrage": it is precisely the statement that the fitted surface is consistent with *some* arbitrage-free dynamic model, even though the fitter never names one.

Heuristic

Convex order with equal means says the longer law is a mean-preserving spread of the shorter: same forward, more dispersion — exactly what a diffusion does as variance accumulates. Enforcing it between independently-fitted expiries is what stops two good slice fits from implying a negative forward variance between them.

2 The LQD-exact enforcement

Proposition 1 (The asset share is the integrated quantile curve)**.** *For an LQD slice with quantile $Q(z)$ in the logit coordinate (Note 01), the integrated upper-quantile curve equals the asset-share integral at the corresponding grid point:*

$$G(\alpha) = \int_\alpha^1 e^{Q(u)}\,\mathrm{d}u = \int_{z_\alpha}^\infty e^{Q(z)}\,\Lambda(z)(1-\Lambda(z))\,\mathrm{d}z = A(z_\alpha).$$

Proof. Substitute $u = \Lambda(z)$, $\mathrm{d}u = \Lambda(z)(1-\Lambda(z))\,\mathrm{d}z$ in the definition of G and compare with Note 01's asset-share integral $A(z)$ — they are the same object. □

So theorem 1(iii) needs no new computation: every built slice already carries $A(z)$ on the shared quadrature grid, and the calendar constraint is an elementwise comparison of two arrays — no quantile inversion, no per-strike call comparison — with constraint points automatically dense in the wings (the grid is uniform in z, not k).

The surface is fitted nearest-to-farthest. The constraint subgrid takes every 25th node (~ 320 points on the 8001-node grid), and each new slice carries the soft squared-hinge slack

$$\sqrt{W_{\mathrm{cal}}}\,\max(A_i(z_m) - A_j(z_m),\,0), \qquad W_{\mathrm{cal}} = \texttt{calendarWeight} = 10^6, \tag{1}$$

keyed on the constraint *z-coordinates*, so a slice calibrated on a coarser optimization grid enforces it by Hermite evaluation, bit-for-bit equivalent to the node-indexed form on the native grid.

3 The model-agnostic enforcement

The SVI and Multi-Core SIV overlays carry no quantile curve, so they use face (iv) directly: a soft hinge on total variance,

$$\sqrt{W_{\mathrm{cal}}}\,\max(w_{\mathrm{near}}(k_m) - w_{\mathrm{far}}(k_m),\,0), \tag{2}$$

on a log-moneyness grid, reading only the previous slice's `implied_w`(k). (The local-vol surface needs *neither* enforcement: a positive local variance is calendar-clean by construction — Note 04's theorem 1(iv) holds along the whole PDE march — so the floor threads only to the parametric overlays.)

The production plumbing is the part that took care. The floor requires the *previous* displayed slice, so `enforceCalendar` threads `prev_display` ascending-T through every path that fits a slice: the single-slice overlay, `fit_and_commit_slice`, the full-surface loop, and the coupled Calibrate workflow all pass the just-fitted display to the next expiry. Wherever the floor is absent (`off`, or the first expiry) the residual block is absent too — byte-identical, test-locked for SVI, SIV, and the display pipeline.

4 Case file: the phantom calendar

Case file: the phantom calendar violation

Setup. A steep short-dated slice over a flatter long-dated one — a single-name skew: near SVI $b = 0.20, \rho = -0.7$; far $b = 0.08, \rho = -0.3$; the far expiry's quotes span only $k \in [-0.25, 0.20]$. On the data, the far slice sits *above* the near one everywhere: genuinely calendar-consistent.
Failure mode. Live SVI overlays on NVDA (Sep-26) and SPY (Jun-27) fitted *flat* with huge RMS whenever the calendar toggle was on.
Diagnosis. The floor (2) was sampled on a fixed wide grid $k \in [-1, 1]$. With *linear* wings, the steep near slice extrapolates to far higher variance at the grid edges than the flat far slice — so $w_{\text{near}} > w_{\text{far}}$ out where neither expiry has a single quote. A *phantom* violation, manufactured entirely by two extrapolation laws disagreeing in a no-data region; the hinge then flattened the far fit to cure a problem the data never posed (fig. 1, left).
Fix. Confinement: `variance_floor_grid_from` builds the floor grid on the observed quote span $[\min k, \max k]$ only (41 points; the wide $[-1, 1]$/161-point grid survives only as the empty-quotes fallback). The two-curve constraint is sampled only where both curves are pinned by data — the confinement principle of Note 09, of which this incident is the founding instance.
Verdict (reproduced with the production calibrator, fig. 1). Under the wide grid the far fit misses its own quotes by 151 vol bp; under the confined grid by 0.0 — indistinguishable from the unconstrained fit. Test-locked, including the assertion that the scenario is consistent on data and inconsistent only in extrapolation.

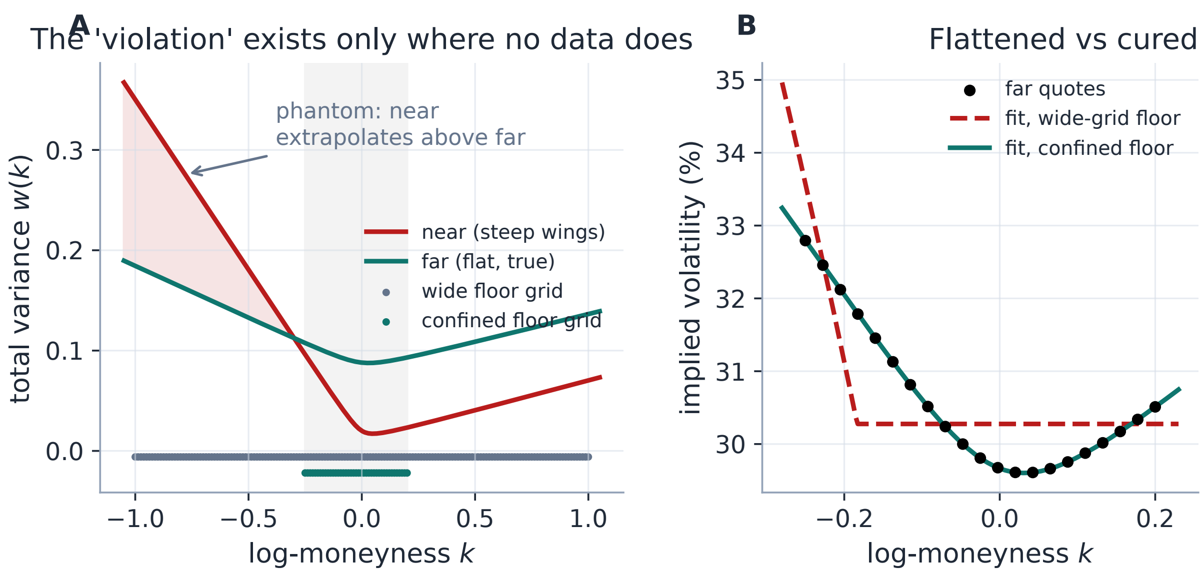


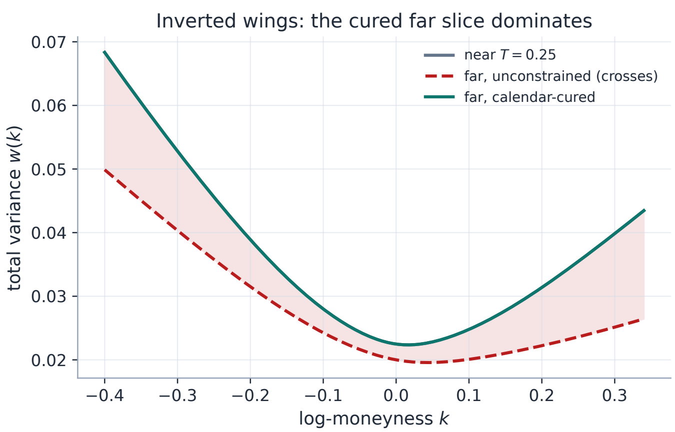
Figure 1: The phantom calendar, reproduced with the production calibrator. Left: on the traded range (shaded) the far slice dominates the near one — no true violation — but the steep near wing extrapolates above the flat far wing exactly where the wide floor grid (dots) samples; the "violation" exists only where no data does. Right: fitted under the wide-grid floor the far smile flattens away from its own quotes (151 vol bp); confined to the traded range it is indistinguishable from the unconstrained fit (0.0 vol bp).

Listing 1: The confinement fix, `variance_floor_grid_from` (verified in appendix B).

```
def variance_floor_grid_from(k_quotes, n=41):
    k = np.asarray(k_quotes, float)      # floor on the TRADED span
        only
    if k.size == 0:
        return variance_floor_grid()     # no data at all: wide
            fallback
    return np.linspace(float(k.min()), float(k.max()), n)
```

5 Worked example: curing a real violation

Confinement removes *phantom* violations; this example shows the floor doing its real job. Figure 2 fits an *inverted-wing* pair: a high-vol short expiry ($T = 0.25$) over a calmer long one ($T = 0.5$), the term structure an imminent event produces. Fitted independently, the long slice's total variance dips below the short one's in the wings — a genuine calendar violation of 0.0059. Refitting under the calendar floor lifts it to dominate everywhere (4.6×10^{-7}, numerically zero), and the integrated upper-quantile curves come out correctly ordered, $G_{\text{far}} \geq G_{\text{near}}$ — theorem 1(iii) made visible.



(a) total variance: crossing, then cured

(b) integrated quantiles, ordered

Figure 2: An inverted-wing pair: the unconstrained long slice crosses the short one (violation 0.0059); the calendar floor cures it (4.6×10^{-7}).

6 What is original here

The convex-order characterization is classical; the contributions are implementation. Recognizing that the LQD asset share *is* the integrated upper-quantile curve (proposition 1) makes the calendar constraint a free elementwise comparison rather than a grid of inverted call inequalities. Extending one calendar treatment to every parametric overlay through the total-variance floor (2) — threaded ascending-T through every calibration path — gives the surface a uniform guarantee regardless of which model draws each slice. And the phantom calendar of section 4 is the incident that crystallized the confinement principle now applied in three other places (Notes 04, 05, 09).

7 Traceability

Table 1: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
LQD integrated-quantile constraint; surface repair	proposition 1, (1)	`calib/calendar.py`, `models/lqd/calibrate.py`	`test_surface.py::test_violating_quotes_get_repaired_when_enforced`
Model-agnostic floor reads prev slice's w	(2)	`calib/calendar.py`	`test_overlay_calendar.py::test_variance_floor_reads_prev_total_variance`
Floor crushes real violations (SVI, SIV)	section 5	`models/svi_jw/calibrate.py`, `models/sigmoid/calibrate.py`	`test_overlay_calendar.py::test_svi_floor_crushes_calendar_violation`, `::test_sigmoid_floor_crushes_calendar_violation`
Byte-identical with no floor / off	invariant 3	`models/svi_jw/calibrate.py`, `models/sigmoid/calibrate.py`	`test_overlay_calendar.py::test_svi_no_floor_is_byte_identical`, `::test_build_display_fit_no_floor_byte_identical`
Phantom scenario: consistent on data, phantom in wings	section 4	`calib/calendar.py`	`test_overlay_calendar.py::test_far_above_near_inside_data_but_not_in_wings`
Confined grid fits clean; wide grid flattens	section 4	`calib/calendar.py::variance_floor_grid_from`	`test_overlay_calendar.py::test_wide_grid_breaks_svi_but_data_grid_does_not`
prev_display threaded ascending-T everywhere	section 3	`api/service.py`, `api/workflow.py`	`test_calibration_workflow.py::test_enforce_calendar_threads_prev_into_parametric_items`, `::test_enforce_calendar_threads_prev_overlay_for_non_lqd`
Coupled surface arbitrage-free	theorem 1	`api/workflow.py`	`test_calibration_workflow.py::test_enforce_calendar_surface_is_arbitrage_free`

A Hyperparameter atlas

Table 2: Calendar-arbitrage hyperparameters.

Knob	Default	Role
Surfaced		
`enforceCalendar`	`true`	Master toggle for calendar coupling (all calibration paths).
`calendarWeight` W_{cal}	10^6	Soft-hinge penalty strength. *Not* the graph's edge-trust `calendarWeight` of Note 14, which shares the name.
Hidden		
`CAL_STRIDE`	25	LQD constraint subgrid stride ($\sim$ 320 points on the 8001-node grid).
`VAR_FLOOR_N_DATA`	41	Confined floor grid resolution (listing 1).
`VAR_FLOOR_KMIN/KMAX`	$[-1, 1]$	Fallback wide grid — empty-quotes only.
`VAR_FLOOR_N`	161	Wide fallback grid resolution.
floor tolerance	0	Hinge slack offset in the floor builders.

Performance. The LQD constraint is an array slice and subtraction on the already-built asset share — effectively free; its Jacobian rows are $-\partial A/\partial\theta$ on the active constraints, supplied by the analytic Jacobian of Note 01. The model-agnostic floor is 41 evaluations of the previous slice's $w(k)$. Confinement *reduces* work by not sampling the extrapolation region.

B Reference implementation

The two enforcement residuals are one line each, and the confined grid is listing 1. The hinge below reproduces the residual blocks inside the SVI/SIV calibrators exactly (the production rows are inline in each calibrator, not a shared function — deliberately, since each model evaluates its own w); listing 1 matches `variance_floor_grid_from` output to machine precision on quoted and empty inputs alike.

Listing 2: The calendar hinge as it appears (inline) in each parametric calibrator, and the LQD asset-share form.

```
# SVI / Multi-Core SIV (face (iv)):    rows are zero unless the far
   dips
cal = np.sqrt(calendar_weight) * np.maximum(cal_floor - w_model, 0.0)

# LQD (face (iii)): the same hinge on the asset share A(z)
cal = np.sqrt(cal_weight) * np.maximum(cal_floor - slice_.
   asset_share_at(cal_z), 0.0)
```

References

[1] V. Strassen. The existence of probability measures with given marginals. *Ann. Math. Statist.*, 36(2):423–439, 1965.

[2] H. Kellerer. Markov-Komposition und eine Anwendung auf Martingale. *Math. Ann.*, 198:99–122, 1972.

[3] P. Carr and D. Madan. A note on sufficient conditions for no arbitrage. *Finance Research Letters*, 2(3):125–130, 2005.

[4] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.